The maximal-perimeter linear-growth question

Exact affirmative answer in later literature

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Status. `literature_already_answered`. This packet identifies and checks an exact later-literature answer; it does not claim a new proof.

Source question

For a log-concave probability measure μ on $\mathbb{R}^n$, let

$$\Gamma(\mu) = \sup_A \mu^+(\partial A), \qquad \Gamma_n = \sup\{\Gamma(\mu) : \mu \text{ isotropic and log-concave on } \mathbb{R}^n\},$$

where A ranges over convex bodies. Brazitikos, Giannopoulos, Hmadi, and Tziotziou, *On the maximal perimeter of isotropic log-concave probability measures*, arXiv:2602.03831, prove

$$cn \le \Gamma_n \le Cn^{3/2}$$

and ask on page 4 whether Γ_n grows linearly with the dimension.

for an absolute constant $C > 0$, where $f_T(x) = f(T^{-1}(x))$ denotes the density of $T_*(\mu)$ and $\|f_T\|_\infty = \|f\|_\infty$. Note that $\Gamma(\mu)$ is not invariant under volume preserving transformations.

If μ is an isotropic log-concave probability measure, then $\|f\|_\infty^{1/n}$ coincides with the isotropic constant of f, which is bounded by an absolute constant (see Section 2). Therefore, (1.4) implies that every isotropic log-concave probability measure μ on $\mathbb{R}^n$ admits a push-forward $T_*\mu$ such that

$$\Gamma(T_*\mu) \leqslant Cn.$$

This result once again raises the natural question of whether Γ_n grows linearly with the dimension.

2 Notation and preliminaries

We work in $\mathbb{R}^n$, equipped with the standard inner product $\langle \cdot, \cdot \rangle$. The associated Euclidean norm is denoted by $|\cdot|$, the Euclidean unit ball by B_2^n, and the Euclidean unit sphere by S^{n-1}. Lebesgue measure in $\mathbb{R}^n$ is denoted by vol_n, and we write $\omega_n = \mathrm{vol}_n(B_2^n)$ for the volume of the Euclidean unit ball. We denote by σ the rotationally invariant probability measure on S^{n-1}.

Throughout the paper, the symbols $C, c, c', c_1, c_2, \ldots$ denote absolute positive constants whose values may change from line to line. Whenever we write $a \approx b$, we mean that there exist absolute constants $c_1, c_2 > 0$ such that $c_1 a \leqslant b \leqslant c_2 a$. All absolute constants are independent of the dimension n.

§2.1. Convex bodies. A convex body in $\mathbb{R}^n$ is a compact convex set K with nonempty interior. It is called symmetric if $K = -K$, and centered if its barycenter $\mathrm{bar}(K) = \frac{1}{\mathrm{vol}_n(K)} \int_K x\,dx$ is at the origin. For every convex body $K \subset \mathbb{R}^n$ we denote by $\overline{K}$ the homothetic copy of K scaled to have unit volume, namely

Exact supporting answer

Eskenazis, Giannopoulos, and Tziotziou, *Functional perimeter and the dimensional Brunn–Minkowski inequality for log-concave measures*, arXiv:2605.02747, explicitly cite arXiv:2602.03831 as the preceding $O(n^{3/2})$ result. Their Theorem 1.2 states

$$\boxed{\Gamma_n \asymp n.}$$

Thus the source question has a full affirmative answer, with both upper and lower bounds of the requested order and with no symmetry assumption beyond the source's isotropic log-concave hypotheses.

$$\mathbf{\Gamma}(\mu) = \sup\{\mu^+(\partial A) : \ A \text{ is a convex body in } \mathbb{R}^n\}.$$

This quantity has been studied for the Gaussian measure by Ball and Nazarov [7, 52] and for rotationally invariant log-concave measures by Livshyts [41, 42, 43]. In a more recent work [44], Livshyts studied the maximal perimeter $\mathbf{\Gamma}(\mu)$ of isotropic measures μ establishing a lower bound which, together with the resolution of the thin shell conjecture by Klartag and Lehec [36] and the result of Nazarov [52], yields

$$\gamma_n = \inf\{\mathbf{\Gamma}(\mu) : \ \mu \text{ is an isotropic log-concave probability measure on } \mathbb{R}^n\} \approx n^{1/4}.$$

For the corresponding supremal quantity

$$\mathbf{\Gamma}_n = \sup\{\mathbf{\Gamma}(\mu) : \ \mu \text{ is an isotropic log-concave probability measure on } \mathbb{R}^n\},$$

an upper bound $\mathbf{\Gamma}_n \leqslant Cn^2$ was established in [44] which was recently improved to $\mathbf{\Gamma}_n \leqslant Cn^{3/2}$ in [12]. We provide an asymptotically sharp bound for $\mathbf{\Gamma}_n$.

Theorem 1.2. *For every $n \in \mathbb{N}$, we have $\mathbf{\Gamma}_n \approx n$.*

Both our main results rely upon the following asymptotically sharp upper bound for the L_1 norm of the gradient of the logarithmic potential (1.5), which was first proven in [22, Lemma 11].

Theorem 1.3 (Eldan and Klartag)**.** *Let μ be an isotropic log-concave probability measure on $\mathbb{R}^n$ with density $f = e^{-\psi}$. Then,*

Proof mechanism in the supporting paper

Let f be the density of an arbitrary isotropic log-concave probability measure and put $K_t = \{x : f(x) \geq t\}$. These level sets are convex. The paper proves the sharp coarea estimate

$$\int_0^\infty \mathcal{H}^{n-1}(\partial K_t)\,dt \leq Cn. \tag{1}$$

For every convex body A, layer cake on ∂A, boundary inclusion, and monotonicity of surface area under inclusion of convex bodies give

$$\begin{aligned}\mu^+(\partial A) &= \int_{\partial A} f\,d\mathcal{H}^{n-1} = \int_0^\infty \mathcal{H}^{n-1}(\partial A \cap K_t)\,dt \\ &\leq \int_0^\infty \mathcal{H}^{n-1}(\partial(A \cap K_t))\,dt \leq \int_0^\infty \mathcal{H}^{n-1}(\partial K_t)\,dt \leq Cn.\end{aligned}$$

Taking the supremum over A yields $\Gamma_n \leq Cn$. The isotropic probability measure on a cube supplies the matching lower bound $\Gamma_n \geq cn$.

Proof of Theorem 1.2. Let f be the density of the isotropic log-concave probability measure μ on $\mathbb{R}^n$, and let $A \subset \mathbb{R}^n$ be an arbitrary convex body. Set

$$K_t = \{x : \ f(x) \geqslant t\}, \qquad t \geqslant 0.$$

We have

$$\mu^+(\partial A) = \int_{\partial A} f(x)\, d\mathcal{H}^{n-1}(x).$$

Applying the layer-cake representation on ∂A, we get

$$\int_{\partial A} f(x)\, d\mathcal{H}^{n-1}(x) = \int_{\partial A} \int_0^{f(x)} dt\, d\mathcal{H}^{n-1}(x) = \int_0^{\infty} \mathcal{H}^{n-1}\left(\partial A \cap \{f \geqslant t\}\right) dt = \int_0^{\infty} \mathcal{H}^{n-1}(\partial A \cap K_t)\, dt.$$

Note that for a closed subset $C \subseteq \mathbb{R}^n$ and an arbitrary subset $M \subseteq \mathbb{R}^n$, we have

$$\partial C \cap M \subseteq \partial(C \cap M).$$

Therefore,

$$\mathcal{H}^{n-1}(\partial A \cap K_t) \leqslant \mathcal{H}^{n-1}(\partial(A \cap K_t)) \leqslant \mathcal{H}^{n-1}(\partial K_t),$$

where the last inequality is the monotonicity of surface area of convex sets with respect to inclusion. Combining the above, we get by (4.1) that

$$\mu^+(\partial A) \leqslant \int_0^{\infty} \mathcal{H}^{n-1}(\partial K_t)\, dt \leqslant Cn.$$

This implies that $\mathbf{\Gamma}(\mu) \leq Cn$, which in turn yields $\mathbf{\Gamma}_n \leq Cn$. The asymptotically matching lower bound follows from considering μ to be the isotropic probability measure on a cube (see also [12, Section 5]). $\square$

Proof intuition

The boundary of a test body can only meet each density level set along a piece of boundary whose surface area is no larger than that of the whole level set. Layer cake sums these contributions over all density levels. The difficult theorem in the later paper is precisely that the total surface area of all levels is $O(n)$ for an isotropic log-concave density. Once that functional-perimeter estimate is available, every convex test body inherits the same $O(n)$ bound at once.

Scope and provenance

The identification is exact rather than merely implied: the later paper names the same quantity, cites arXiv:2602.03831 for the preceding bound, and states and proves the sharp order. The source question occurs on page 4; the later statement is Theorem 1.2 on page 3, and its short deduction is on page 19.

The two bundled PDFs were reproducibly compiled, without source edits, from the run's archived arXiv source downloads. Their archive SHA-256 values are `1eb7475c...d1339a` (2602.03831) and `628e6a58...ec5a90` (2605.02747). Exact rendered crops of the source question, answer theorem, and proof are included above.

References. [1] S. Brazitikos, A. Giannopoulos, A. Hmadi, and N. Tziotziou, *On the maximal perimeter of isotropic log-concave probability measures*, arXiv:2602.03831 (2026). [2] A. Eskenazis, A. Giannopoulos, and N. Tziotziou, *Functional perimeter and the dimensional Brunn–Minkowski inequality for log-concave measures*, arXiv:2605.02747v2 (2026), Theorem 1.2 and Section 4.